GNEG590V Assignment # 3

1. 2-19, 2-22, 2-26 (Hint: modulation alphabet size for QAM-64: 64), 2-50
2. Compare packet switching with circuit switching.
3. A communication system using Time Division Multiplexing. Each $125\mu s$ period is defined as a frame, and one frame is evenly divided into 8 slots. Each slot is assigned to one user. Each slot contains 162 symbols, and the modulation alphabet size used in the system is 8. Among the 162 modulation symbols, 6 of them are used of control, and the remaining 156 symbols are used for user data. Calculate
   (a) the gross data rate (counting both control and user data) of one user;
   (b) the net data rate (counting only user data) of one user;
   (c) the gross data rate of the entire system (counting the information of all users);
   (d) the net data rate of the entire system.